

EDUCATION

- **Radboud University** Nijmegen, The Netherlands
PhD position in: Calculus of Variations; *Supervisor:* Riccardo Cristoferi
September 2023 - August 2027
Area of interest: Calculus of variations, geometric measure theory, Gamma-convergence, material science and related fields.
- **Polytechnic University of Turin** Turin, Italy
Master degree in: Engineering Mathematics
September 2021 - July 2023
Thesis: Anti-plane analysis of Francfort-Marigo model for quasi-static brittle fracture; *Supervisor:* Marco Morandotti.
- **KTH Royal Institute of Technology** Stockholm, Sweden
Exchange semester
August 2022 - January 2023
- **Polytechnic University of Turin** Turin, Italy
Bachelor degree in: Mathematics for Engineering
September 2018 - July 2021
Thesis: Fractal dimension and Hausdorff measure; *Supervisor:* Luca Lussardi
- **L.S. Battaglini** Taranto, Italy
Scientific Diploma
September 2013 - July 2018

PUBLICATIONS

- **Low regularity potentials in heterogeneous Cahn–Hilliard functionals**
R. Cristoferi, J. Deutsch, L. Pignatelli
Arxiv: 2603.25369
- **Phase separation in multiply periodic materials with fine microstructures**
R. Cristoferi, L. Pignatelli
Arxiv: 2506.14462
- **Phase separation in multiply periodic materials with fine microstructures: a second regime**
L. Pignatelli
ongoing
- **Sharp interface dynamics in heterogeneous domains**
R. Cristoferi, L. Gancedi, L. Pignatelli
ongoing
- **Convergence of critical regime to sub- and super-critical regimes**
R. Cristoferi, L. Pignatelli
ongoing

TALKS

- **Why Soap Films are smarter than Calculus**
16-12-2025, PhD Colloquium
Radboud University, Nijmegen
- **Phase separation in multiply periodic materials with fine microstructures**
10-07-2025, Workshop Calculus of Variations NL
Roermond
- **Phase separation in heterogenous media**
28-05-2025, Seminar on Calculus of Variations
TU Wien, Vienna
- **Homogenization and phase separation in heterogeneous media**
20-03-2025, Seminar talk
RWTH Aachen University, Aachen
- **Gamma Mia - an introduction to Gamma convergence**
14-10-2024, PhD Colloquium
Radboud University, Nijmegen
- **Quasi-static brittle fracture: analysis of Francfort-Marigo model**
22-11-2023, Applied Analysis Seminar
Radboud University, Nijmegen

POSTERS

- **Phase separation in heterogeneous media**
12-06-2025, Mathematical Analysis and Applications
Como
- **Phase separation in multi-scale media**
11-04-2024, Mathematical Advances for Complex Materials with Microstructures
University of Würzburg, Würzburg

RESEARCH VISITS

- **TU Wien, Vienna** 26-05-2025 - 30-05-2025
Collaboration with Jakob Deutsch
- **RWTH Aachen University, Aachen** 17-03-2025 - 20-03-2025
Collaboration with Likhith Ganedi

PROJECTS

- **Reading Group organizer** The Netherlands, September 2023 - ongoing
Series of meetings for the CalcVarNL community - Main topics: Gradient Flows, Optimal Transport, Stochastic Analysis.
- **Modeling Camp organizer** Radboud University, February 2024, February 2025
Collaboration with dutch companies, where problems from the industry are modeled mathematically and numerically.
- **Head of YSC committee** Rabdoud University, March 2024 - ongoing
Young Scientist Council of the Institute for Mathematics, Astrophysics and Particle Physics.

EVENTS ATTENDED

- **Modern Methods in Calculus of Variations** Hausdorff Center for Mathematics
23-02-2026 - 27-02-2026 Bonn, Germany
- **Workshop Calculus of Variations NL** Roermond, The Netherlands
07-07-2025 - 11-07-2025
- **Analysis of Partial Differential Equations** Czech Academy of Sciences
16-09-2024 - 19-09-2024 Prague, Czech Republic
- **Workshop Calculus of Variations NL** Schiermonnikoog, The Netherlands
08-07-2024 - 12-07-2024
- **Vito Volterra Meeting in Calculus of Variations 2024** Sapienza University of Rome
24-06-2024 - 28-06-2024 Rome, Italy
- **Mathematical Advances for Complex Materials with Microstructures** University of Würzburg
08-04-2024 - 12-04-2024 Würzburg, Germany
- **Variational models for material failure** FAU Erlangen-Nürnberg
20-09-2023 - 22-09-2023 Erlangen, Germany
- **Analysis of Partial Differential Equations** KU Eichstätt-Ingolstadt
18-09-2023 - 20-09-2023 Eichstätt, Germany
- **Beyond Elasticity: Advances and Research Challenges** CIRM
16-05-2022 - 20-05-2022 Luminy, France

TEACHING

- **Numerical Methods for PDEs** February 2026 - June 2026
TA for the bachelor course Numerical Methods for PDEs prof. V. Nikolić
- **Numerical Methods** September 2025 - January 2026
TA for the bachelor course Numerical Methods prof. L. Scarabosio
- **Calculus of Variations** February 2025 - June 2025
TA for the MasterMath course Calculus of Variations prof. A. Bach, prof. R. Cristoferi
- **Multivariable Analysis** September 2024 - January 2025
TA for the bachelor course Multivariable Analysis prof. R. Cristoferi
- **Numerical Methods for PDEs** February 2024 - June 2024
TA for the bachelor course Numerical Methods for PDEs prof. V. Nikolić
- **Analysis 2** September 2023 - January 2024
TA for the bachelor course Analysis 2 prof. R. Cristoferi

SKILLS

- **Languages:** Italian (native), English (fluent), Dutch (basics), Spanish (intermediate)
- **Coding:** C++, Python, Matlab, L^AT_EX– all advanced

ACADEMIC REFEREES

- **Riccardo Cristoferi:** Radboud University, cristoferi@science.ru.nl
- **Marco Morandotti:** Polytechnic University of Turin, marco.morandotti@polito.it

Luca Pignatelli
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15-04-2026